Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
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Linked list

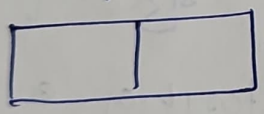
* The everday usage of the term "list" refers to a linear collection of data item.
* one way to store data in linear way is ~~list~~ Array, But array has certain disadvantage - e.g. it is relatively expensive to insert and delete element in an array. Also since array occupies a block of memory space, one cannot double or ~~triple~~ the size of an array
* Another way of storing a list in memory is to have each element in the list contain a address field which contain the address of next element in the list.

This data structure also called Linked list.

Linked list -

A linked list or one way list is a linear collection of data elements called nodes, where the linear order is given by means of pointers. That is, each node is divided into two parts; the first part contain the information of the element, and the second part called the link field or next pointer field contain the address of the next node in the list

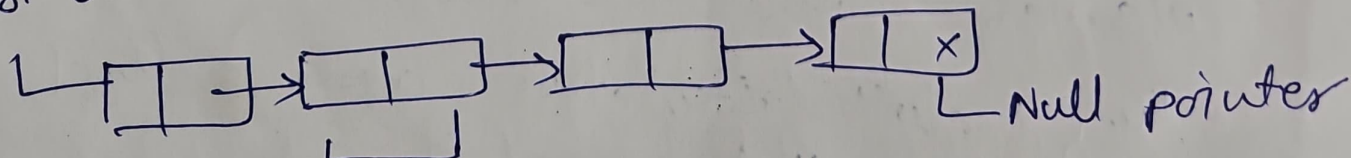
```
struct Node
{ int info;
  Node *next;
} *start;
```



↳ next pointer field

Information part

start or Head



Linked list

	Array	Linked list
③ Chang of Size	Static in nature	Dynamic (may extend or shrink)
④ Inserting Deletion	Insertion / Deletion opn are slow, half of elements need to arange during opn	I/D opn are perform very fast in constant amount of time
⑤ Searching	Linear Search Binary Search interpolation Search are possible	Binary search, inter ~~postit~~ Polation Search not possible only linear Search
Access element	fast access in constant Amt of time	To access any element traversing is required.

Types of linked list

- ☆ Singly linked list
- ☆ Circular linked list
- ☆ Doubly linked list
- ☆ Header linked list
- ☆ circular Doubly LL

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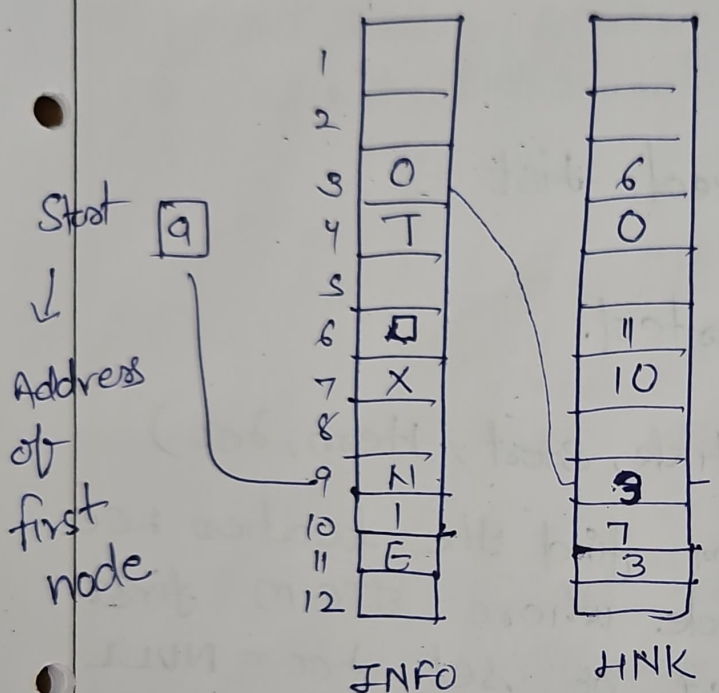
Application - * used in dynamic memory allocation,
* used in implementation of stack, queue, graph, trees
* used in music player, web browsing to store URL of previous & next page, image, viewer, etc.

Difference

	Array	Linked list
Storage	Storage of array must be available contigously.	Need not be contigous.
Memory utilization	Size need to be predicted beforehand	can be allocated wherever required
	No extra space can't be possible, nor shared with any other free space	If required extra space for storing required for address address

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Representation of linked list in memory



operation on linked list

① **Traversing:** LIST – linked list, Ptr → temporary variable

1. set Ptr = Start;
2. Repeat step 3 and 4 while Ptr ≠ NULL
3. Apply Process to Info[Ptr]
4. set Ptr = Link[Ptr]
5. Exit

Algorithm to count total no. of nodes in given list.

(2) S

(2) Searching A linked list

(1) list is unsorted.

· search (info, link, start, item, loc)

// This algorithm find the location LOC of the node where ITEM first appears in list or set LOC = NULL

1. set PTR = start
2. Repeat step 3 while PTR ≠ NULL
3. If ITEM = INFO[PTR], then
   set LOC := PTR and exit
   Else
   set PTR = LINK [PTR]
   End of If
4. [Search unsuccessful] Set LOC = NULL
5. Exit

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(2) When list is sorted.

SEARCH_SORT_LIST (INFO , LINK, START, ITEM, LOC)

1. Set PTR = START
2. Repeat Step 3 while PTR ≠ NULL
3. If ITEM < INFO[PTR] then
   Set PTR = LINK[PTR]
   Else if ITEM = INFO[PTR] then;
   Set LOC = PTR and Exit
   Else
   Set LOC = NULL and Exit
   [item now exceeds INFO[PTR]]
   End of If
   End of step 2 loop
4. Set LOC = NULL.
5. Exit

100	75	45	25	15	5

ITEM = 35

Binary search can't be applied on a
So, the worst case running time is proport
to number n of element in HST.
and Average case running time is n/2.

Memory Allocation & GARBAGE Collection

together with the linked list in memory, a speaical list is maintained which consist of unused memory cells. This list has its own pointer; is called the list of available space or free storage list.

HST(INFO, HNK, START, AVAIL)

Garbage collection - Any technique which Handle the free storage or deleted element space collection is called garbage collection Generally, the garbage collection is invisible to the progeammer.

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Overflow and Underflow -

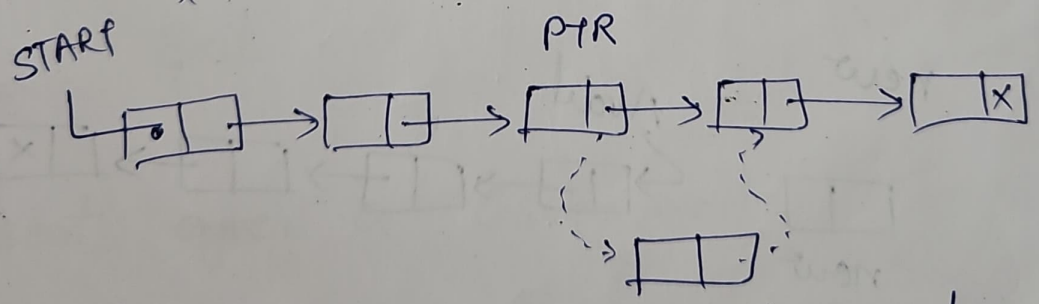
When we try to insert a new data but there is no available space, This situation is usually called overflow.

[when Avail == NULL, → overflow]

Underflow refers to the situation where one wants to delete data from a data structure that is empty.

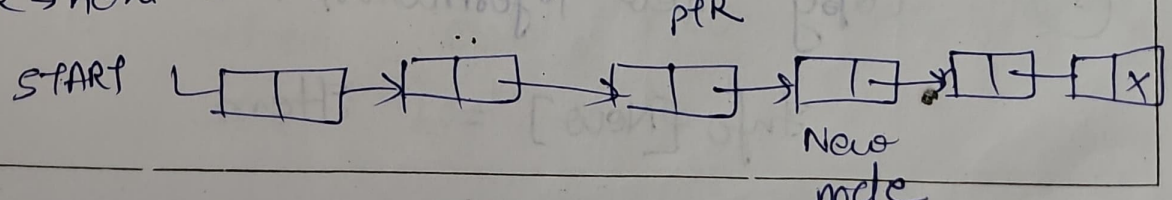
[when Start = NULL, — Underflow]

Insertion into a linked list



New → next = PTR → next

PTR → next = new



Insertion can be possible in 3 way

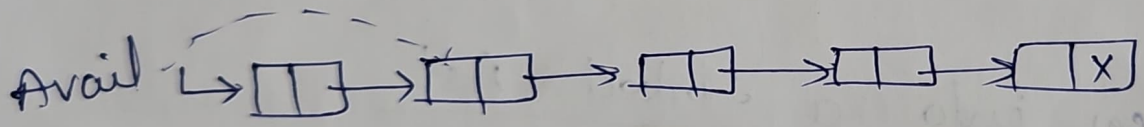
① Insert at the Beginning of a list
② Insert at the END
③ Insert after a Given Node

following steps will include :

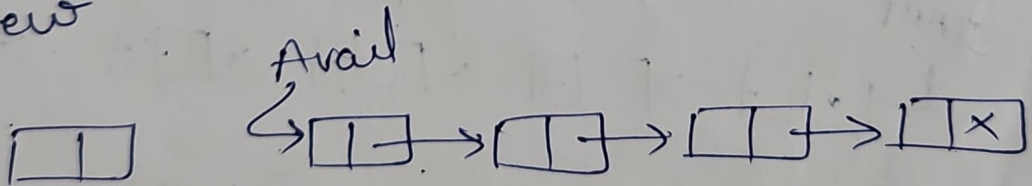
① checking if space is available or overflow condition

② Remove first node from the availe list

New = Avail
Avail = Link[Avail]



new



③ Copy new information into the new node

Info [New] = Item

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Algo ①

InsertFirst (Info, link, Start, Avail, Item)

1. [Overflow] If Avail = NULL, then
   write "OVERFLOW" and Exit
2. [Remove first node from Avail list]
   New = Avail;
   Avail = link[Avail]
3. Set Info[New] = Item;
4. link[New] = START
5. START = NEW [new pointer to start]
6. Exit.

Insert after a Given Node

① Traverse upto End of the list
② then add new node to the last node

Algo ②

Insert_at_END(INFO, LINK, START, Avail,

1. [overflow ?] If Avail == NULL then overflow and Ex

2. [Remove first node from avail]

   ~~Avail~~ new = Avail

   Avail = LINK[Avail]

3. Traverse the list until reach at End

   PTR = START ;

   while (PTR ≠ NULL)

   {

   LOC = PTR

   PTR = PTR → next

   }

4. [LOC hold the last node]

   set next[LOC] = New

5 Next[New] = NULL

6. Exit

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③ Inserting after a Given Node

INSLOC(INFO, LINK, START, AVAIL, LOC, ITEM)

1. [Overflow ?] If Avail == NULL then write "OVERFLOW" and Exit
2. [Remove first node from Avail list]
   New = Avail and
   Avail = LINK[Avail]
3. Set Info[NEW] = Item
4. [Insert at first, If LOC = NULL, START=NULL]
   IF LOC = NULL then:
   Set LINK[NEW] = START
   START = NEW
5. Else:
   Link[NEW] = Link[LOC] and
   Link[LOC] = New
   [End of If structure]
6. Exit.

(4) Inserting into a Sorted list

a → search for
b → Insert at loc

Search Loc(Info, link, start, Item, loc)

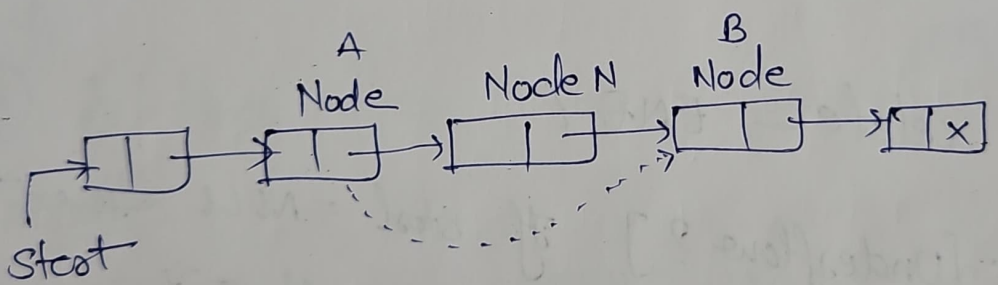
1. [list empty?] If start = NULL then set
   Loc = NULL and Return
2. [special case?] If Item < Info[start]
   then set Loc = NULL and Return
3. set save = start and
   ptr = link[start]
4. Repeat step 5 and 6 while PTR ≠ NULL
5. If Item < Info[PTR], then:
   set Loc := save and Return
   End of If
6. set save = PTR and
   PTR = link[PTR]
   End of loop
7. set LOC = SAVE
8. Return

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Insert (Info, Link, Start, Avail, Item)

1. call Search-Loc (———)
2. call Insert-Loc (———)
3. Exit

Deletion from a linked list



Link [NodeA] = Link [NodeN]

Now Add the NodeN in Avail list

Link [Avail] = NodeN

Avail = NodeN

Algo.

(1) Delete_BEG (Info, Link, Start, Avail, ter)

1. [Underflow ?] If Start = NULL then
   Print "Underflow" and Exit
   temp = ~~Loc~~ start
2. Set Start = Link[Start]
3. Add the deleted node in Avail
4. ~~Link[Avail] temp~~.

   Link[temp] = Avail
   Avail = temp.

(2) Delete_END (.

1. [Underflow ?] If Start = NULL then
   Print "underflow" and Exit
2. [Traverse the list upto last]
   set Ptr = Start
3. [Repeat step ___ while Link[Ptr] ≠ NULL]
   set P.Ptr = Ptr
   Ptr = Link[PTR]
4. Delete last Node

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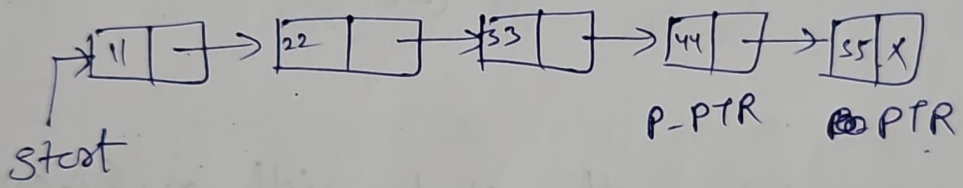
Link[P-Ptr] = NULL

5 Add deleted node in Avail list

Link[PTR] = Avail

Avail = PTR

6. Exit



(3) Deleting the Node following a Given Node

Algorithm

Delete(Info, Link, START, Avail, LOC, LOCP)

1. If LOCP = NULL then:

set start = Link[START]

[Delete first node]

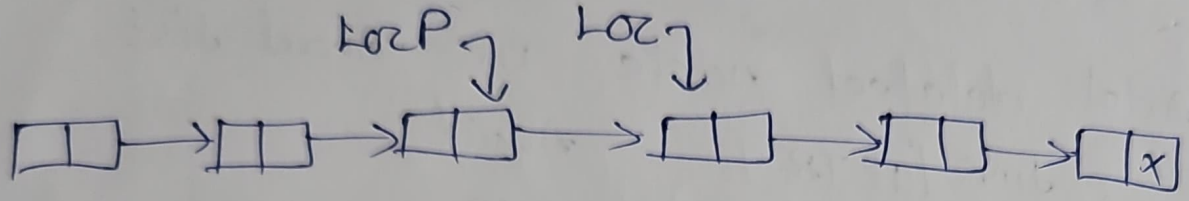
Else

set Link[LOCP] = Link[LOC]

[Delete node N]

End of If

2. [Return deleted node]
   Link [LOC] = Avail and
   Avail = LOC
3. Exit.



Link [LOCP] = Link[LOC]

Algo Deleting the Node with a Given Item of Information

LOC – Location of node N which contain Item
LOCP – Preceding node location of Node N
If Item is not in the list then LOC = NULL
If Item is at first node, then LOCP = NULL

1. [List empty?] If START = NULL then
   Set LOC = NULL and LOCP = NULL and
   Return

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2. [Item in first node] If Info[START] = ITEM then LOC = START and LOCP = NULL & Ret

3. Set SAVE = START and PTR = LINK[START]

4. Repeat step 5 and 6 while PTR ≠ NULL

5. If Info[PTR] = Item then
   set LOC = PTR and LOCP = SAVE
   and Return

6 Set SAVE = PTR and PTR = LINK[PTR]
End of step 4 loop

7 set LOC = NULL (search unsuccessful)

8 Return

Delete (Info, Link, START, Avail, Item)

1. Call FindB(_____)

2. If LOC = NULL then write "Item not in list and Exit".

3 [Delete node]
If LOCP = NULL then
set START = Link [START]
Else
set Link [LOCP] = Link [LOC]
End of If structure.

4. [Return deleted node to Avail list
Link [LOC] = Avail and
Avail = LOC

5 Exit

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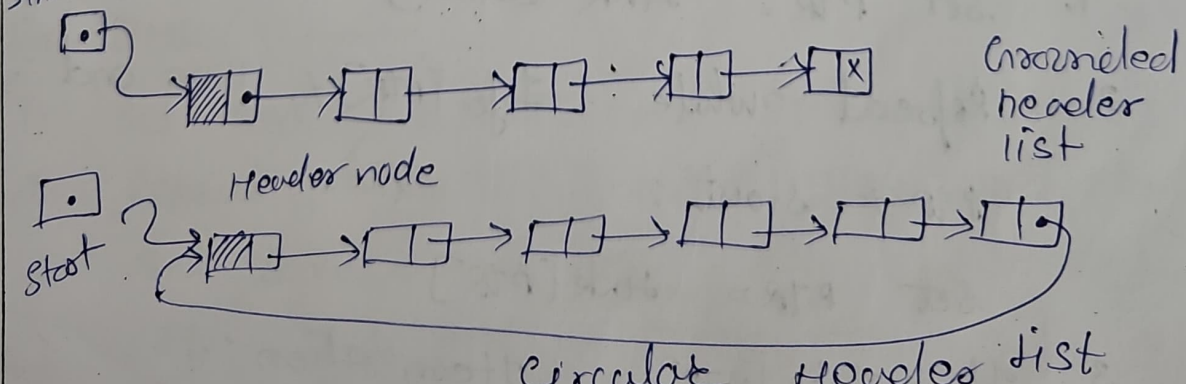
Header linked lists

A header linked list is a linked list which always contains a special node, called the header node, at the beginning of the list.

Two kinds of widely used Header list

(1) A grounded header list is a header list where the last node contains the null pointer.

(2) A circular header list is a header list where the last node points back to the header node.



Properties of circular header list

① The null pointer is not used, and hence all pointer contain valid address

② Every node has a predecessor

Traverse (Info, Link, Head)

1. set PTR = Link[start]
2. Repeat step 3 & 4 while PTR ≠ START
3. Apply Process to Info[PTR]
4. set PTR = Link[PTR].
5. Exit

② SEARCH (Info, Link, Start, Item, LOC)

1. set PTR = Link [start]
2. Repeat while Info [PTR] ≠ Item and PTR ≠ start :
   set PTR = Link[PTR]
3. If Info [PTR] = Item then :
   set LOC = PTR

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Else
set loc = NULL
4. Exit

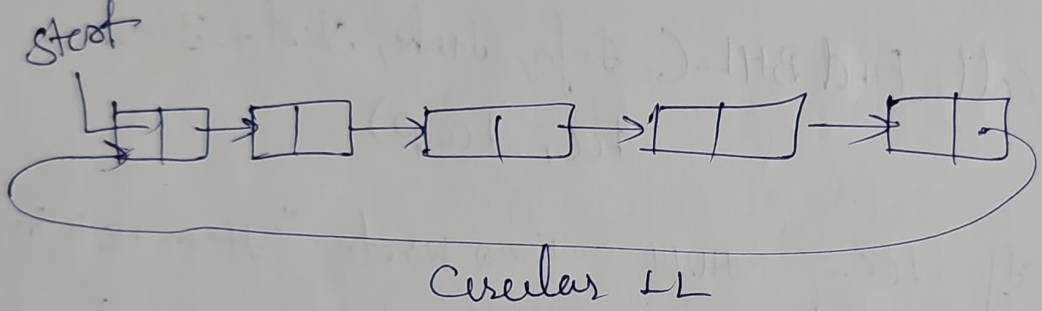
Delete
1. Call FindBHL(Info, link, start, Item, loc, locp)
2. If loc = NULL then write "Item not in list" and exit.
3. Set link[locp] = link[loc]
4 link[loc] = Avail and Avail = loc
5 Exit

FindBHL (———)
1. set SAVE = START and PTR = link[start]
2= Repeat while Info[PTR] ≠ Item and PTR ≠ Start
set Save = PTR and PTR = link[PTR]
3. If Info[PTR] = Item then
set loc = PTR and locp = Save
else loc = NULL and locp = Save
4 Exit

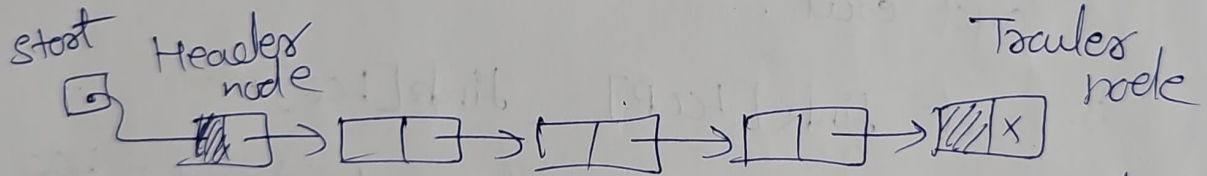
Two other variations of linked list

① A linked list whose last node points back to the first node called circular list

② A linked list which contains both a special header node at the beginning of the list and trailer node at the end of the list



Circular LL



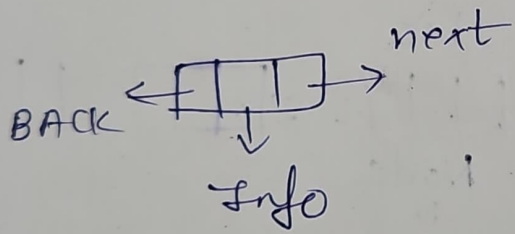
LL with Header & Trailer nod

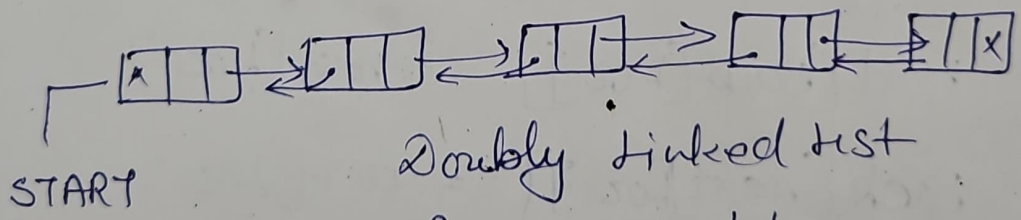
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Two-Way-List-

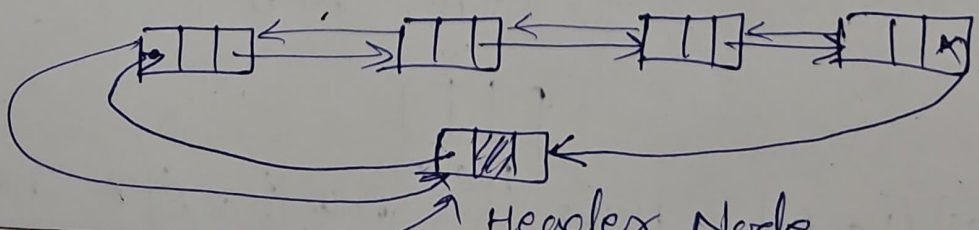
A two way list is a linear collection of data elements, called nodes, where each node N is divided into three parts

1. Info
2. NEXT
3. BACK or PREV





Doubly linked list
or two way LL



two way HLL

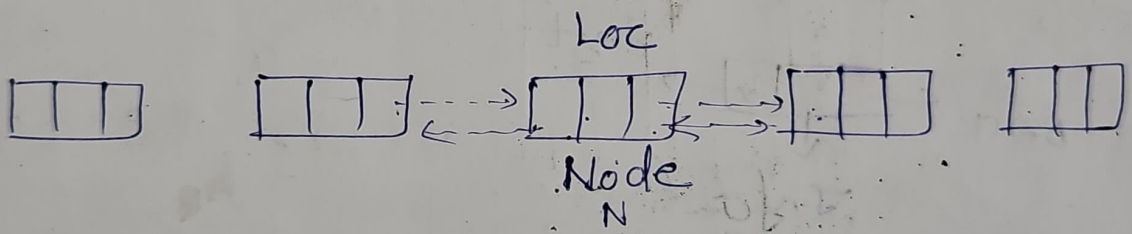
Operation on two way list

① Traversing < same as SLL
No advantage

② Searching → Same as SLL but can start at end as well

→ If list is sorted, we can start search at End

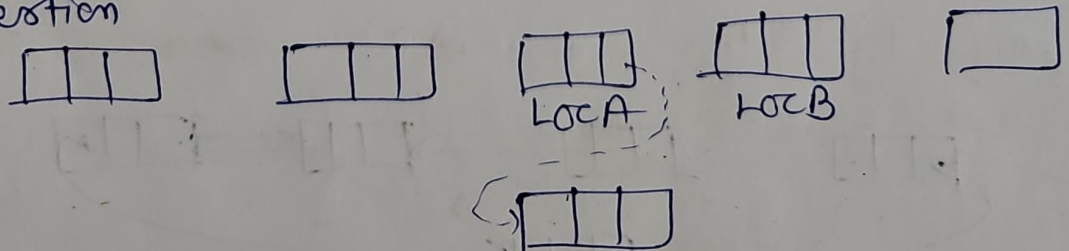
③ Deleting - Advantage - In Two way list No need to travel the whole list for finding the location of previous node.



next[Back[Loc]] = next[Loc]

Back[next[Loc]] = Back[Loc]

④ Insertion



next[LocA] = new
Back[New] = LocA

Back[LocB] = new
Next[New] = LocB